

SIMATS ENGINEERING

Co3-ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyse the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Dr.V. Parthipan

QUESTIONS: 4 Total Marks: 40

Code of Conduct:

I **B Tharun Kumar/192424356** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *B Tharun Kumar*

Criteria	Understanding the case 2.5	Application of Concepts 2.5	Analysis 2	Solution Design 2	Clarity 1	Total (10)
Q1						
Q2						
Q3						
Q4						

Faculty Signature: _____

Question 1 – Real-Time Video Conferencing Degradation

Case Scenario: *A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.*

1.1 Analysis: Transport Protocol or Application Layer?

Layer identification: The metrics reported — 8% packet loss, 180 ms jitter, and fluctuating bandwidth — are transport/network-layer metrics, not application-layer metrics such as CPU load or rendering delay. This points to the transport layer as the primary source of the problem.

If TCP is used: TCP's reliability mechanism forces in-order delivery, so a single lost segment blocks all later segments until it is retransmitted (head-of-line blocking). This produces exactly the frozen-frame and delayed-audio symptoms described in the case.

If UDP is used: UDP does not block on loss — the decoder simply skips or conceals missing data. Complete freezes (rather than minor glitches) suggest either a TCP fallback path is active, or the UDP jitter buffer/FEC is poorly tuned for 180 ms of jitter.

Conclusion: The root cause is transport-layer behaviour under loss and delay. Application-level buffering is a secondary, contributing factor: a well-tuned adaptive buffer can mask some transport impairment, a fixed/shallow buffer cannot.

1.2 Justification Based on Real-Time Multimedia Requirements

Timeliness over completeness: File transfer values correctness over speed, so TCP's retransmit-and-reorder model fits it well. Live video/voice values timeliness over completeness — a late frame is useless even if eventually delivered, since a live conversation cannot pause and wait.

Delay budget: Interactive voice should stay under ~150 ms one-way (ITU-T G.114). TCP retransmission can add hundreds of milliseconds per lost segment, breaking this budget outright.

Jitter tolerance: 180 ms jitter is very high against a normal target of under 30 ms, indicating congestion/queuing on the path or an under-provisioned receive buffer.

Loss tolerance: Voice/video codecs can tolerate roughly 1–5% loss through concealment; TCP instead treats any loss as an error requiring correction — the wrong trade-off for real-time media.

Industry practice: This is why RTP/UDP (as used by Zoom, Teams, Meet, WebRTC) is the default for conferencing, with TCP used only as a firewall-driven fallback.

1.3 Recommended Protocol-Level and Application-Level Improvements

Protocol level: Carry media over RTP/SRTP on UDP with RTCP feedback enabled. Open the required UDP port ranges on the firewall so the platform is not forced onto a TCP fallback. Where UDP is impossible, prefer QUIC/SCTP over plain TCP, since both avoid strict head-of-line blocking.

Application level: Use an adaptive jitter buffer sized to the measured 180 ms jitter, enable forward error correction (FEC) and loss concealment, and use NACK-based retransmission only for keyframes. Enable adaptive bitrate/resolution (simulcast or SVC) so the encoder backs off automatically when bandwidth drops.

Network level: Apply DSCP/QoS marking (EF for voice, AF41 for video) across the internal network and WAN links so real-time media is prioritised over background data traffic during peak hours.

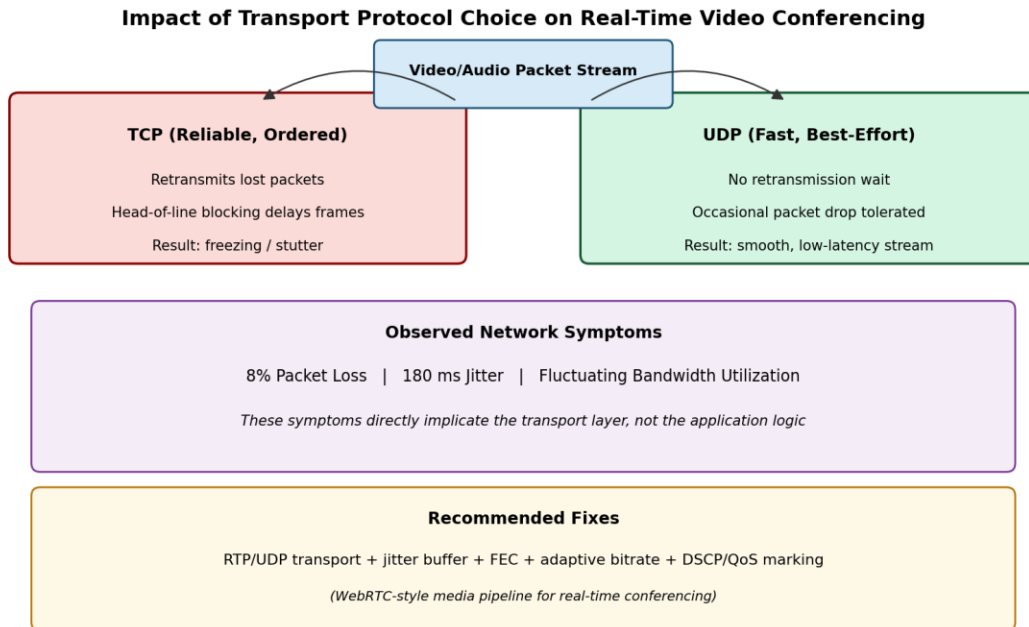


Figure 1.1: Why UDP-based real-time transport outperforms TCP for live video conferencing under packet loss and jitter.

Question 2 – High DNS Resolution Time (800 ms Homepage Delay)

***Case Scenario:** A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.*

2.1 Analysis: Reasons Behind the High DNS Resolution Time

Scale of the problem: 800 ms is roughly 40-80x higher than a well-designed cached DNS setup, which typically resolves in 5-20 ms.

Geographic distance: If every query worldwide reaches one centrally located server, employees far from that server pay the full long-haul round-trip cost, often across several referral hops.

Poor caching: Low TTLs or region-specific uncached resolvers force a full iterative resolution (root → TLD → authoritative) on nearly every lookup instead of serving from cache.

Centralisation and retries: A small number of centralised authoritative servers adds latency on the final referral hop, and any timeout/retry against an overloaded or unreachable server is folded into the reported average.

2.2 Proposed DNS Architecture to Reduce Lookup Time Below 50 ms

Anycast DNS: The same DNS IP is announced from multiple regional points of presence (e.g., North America, Europe, Asia-Pacific). BGP automatically routes each query to the nearest node, cutting round-trip distance and giving automatic failover.

DNS pre-fetching: Browsers/applications resolve likely-needed domains in advance, in parallel with page load, removing DNS resolution from the user-visible critical path.

TTL optimisation: Stable records (e.g., the main app domain) get longer TTLs so repeat lookups are served from cache; only records needing fast failover keep short TTLs.

Regional resolvers: Deploying caching resolvers close to each office, paired with the Anycast authoritative layer, keeps both the resolver hop and the authoritative hop geographically short — enabling the sub-50 ms target.

2.3 Advantages and Limitations of the Proposed Architecture

Performance gain: Anycast removes the distance penalty and caching/pre-fetching removes repeated full resolutions, so most queries are answered from cache or a single short round trip.

Scalability: Anycast scales horizontally — new regions can be added without any change to how clients address the service, since load spreads naturally across nodes.

Availability: Automatic failover plus multiple authoritative servers per region means a single node failure degrades only nearby users, not the whole service.

Limitations: BGP convergence during failover is fast but not instant; long TTLs slightly delay propagation of urgent record changes; pre-fetching adds a little unnecessary query traffic; and the distributed footprint costs more to operate than one centralised server.

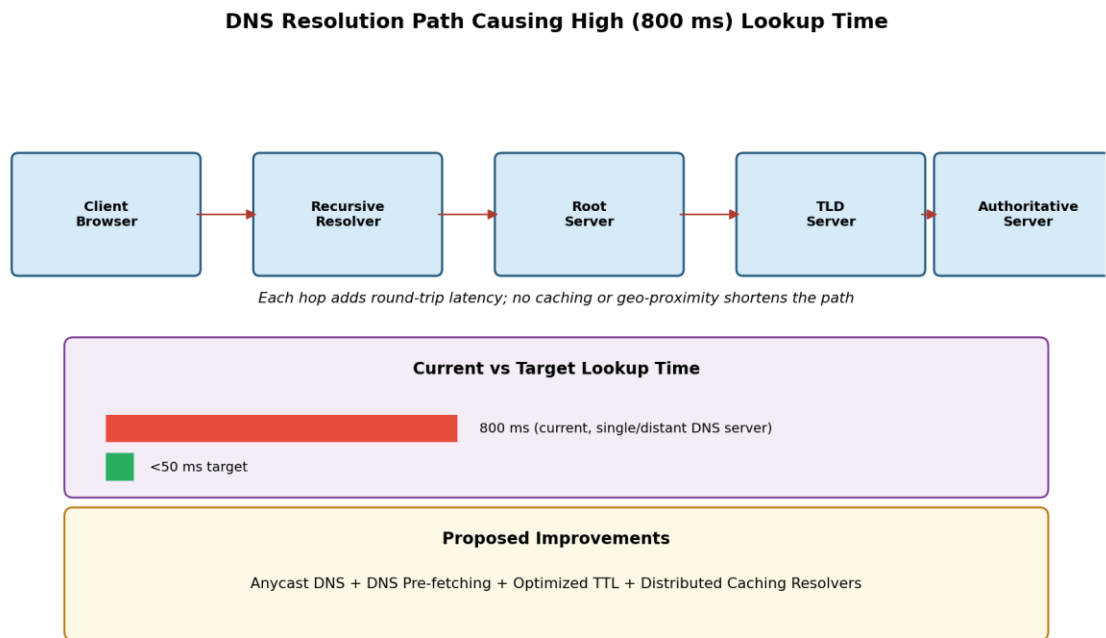


Figure 2.1: Multi-hop DNS resolution path contributing to the 800 ms lookup delay, and the target architecture.

Question 3 – Distributed DNS Solution for Global Cloud Traffic

Case Scenario: *A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.*

3.1 Impact of Centralized DNS Architecture on Application Performance

Latency impact: Every query worldwide must reach one physical server, so users far from it always pay the full long-distance round trip — routing has no way to shorten that distance.

Capacity impact: A single server has a hard queries-per-second ceiling. During promotional-event traffic spikes, queries queue, retry, or time out once that ceiling is approached — explaining the intermittent (not constant) nature of the delay.

Availability impact: A single primary server is a single point of failure: hardware fault, maintenance, or a DDoS attack takes down DNS resolution for the entire global user base at once, since no alternate path exists.

3.2 Distributed DNS Solution Design

Layer 1 – Anycast authoritative clusters: Multiple regional clusters (North America, South America, Europe, Africa/Middle East, Asia-Pacific) all answer under the same Anycast IP; BGP automatically sends each query to the nearest cluster.

Layer 2 – Replica servers: Secondary authoritative servers continuously sync from a hidden primary via zone transfers, so every region serves current data without needing a single point of contact.

Layer 3 – Caching resolvers: Edge and ISP recursive resolvers, tuned with per-record TTLs, absorb repeated queries for the same popular domains — exactly the pattern seen during a promotional spike.

Resilience layer: Continuous health checks automatically withdraw failed nodes from routing within seconds; rate limiting and DDoS-scrubbing protect each point of presence from regional overload.

3.3 Influence of TTL, Anycast Routing, and DNS Caching on Performance and Fault Tolerance

TTL: Longer TTLs cut authoritative-layer query volume sharply, which matters most during a spike; shorter TTLs on critical failover records trade some of that gain for faster propagation of urgent changes.

Anycast routing: Improves performance by answering from the nearest node, and improves fault tolerance because losing any one node simply shifts traffic to the next-nearest healthy node.

DNS caching: Acts as a load-absorbing buffer between end users and the authoritative layer, so only a small fraction of total queries ever reach an authoritative server — reducing both load and the blast radius of any failure.

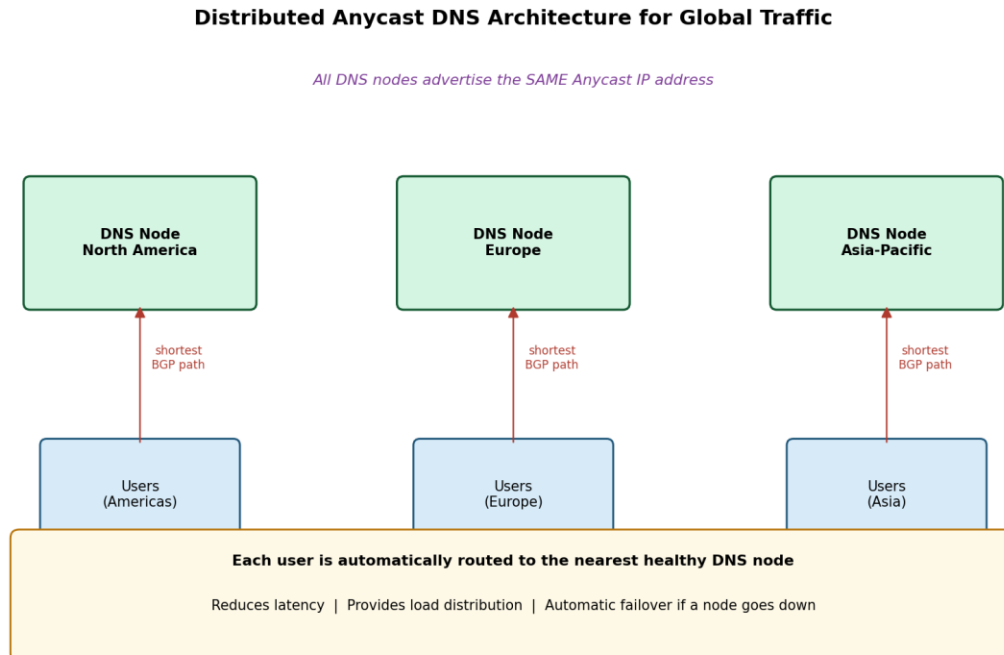


Figure 3.1: Anycast-based distributed DNS architecture routing each user to their nearest regional node.

Question 4 – TCP Latency in a High-Frequency Stock Trading Platform

***Case Scenario:** A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.*

4.1 Three TCP-Related Factors Behind the Latency Increase

Overview: The three reported symptoms — retransmissions, longer connection setup, and excessive buffering — map directly onto three TCP behaviours: flow control, Nagle's algorithm, and SYN queue limits, all amplified by the market-open burst.

Flow control: The receiver's advertised window shrinks (or collapses to zero) when the application/socket buffers cannot drain incoming data fast enough during the surge, forcing the sender to pause and wait.

Nagle's algorithm: Small order/ACK packets are held and coalesced before sending. Combined with delayed ACKs on the receiver side, this interaction can add tens of milliseconds of stall even on a fast network path.

SYN queue limits: The fixed-size backlog for half-open connections overflows when many clients connect simultaneously; excess SYNs are dropped, forcing timeout-and-retransmit with exponential backoff.

4.2 How Each Factor Affects Transaction Processing

Business impact of latency: Every added millisecond means the client's view of order acceptance and price becomes stale — critical in a market that can move within milliseconds.

Flow control impact: A shrinking window stalls every order in the pipeline at that moment, not just one client, which matches the case's report of a platform-wide average latency increase.

Nagle's algorithm impact: It specifically penalises the small, frequent packets that order messages are made of, and — multiplied across thousands of orders per second — is enough on its own to move latency from 1 ms toward 40 ms.

SYN queue impact: Clients that cannot complete the handshake cannot trade at all during that window — a direct business cost during the most active minutes of the trading day, and retries from many clients can worsen the congestion further.

4.3 Recommended TCP Configuration Changes

For flow control: Increase TCP receive/send buffer sizes and enable window scaling (RFC 1323); ensure the application has enough thread/process capacity to drain the socket buffer quickly so the window rarely needs to shrink.

For Nagle's algorithm: Set `TCP_NODELAY` on latency-sensitive order/ACK sockets to disable Nagle's algorithm, and consider `TCP_QUICKACK` on the receiver to remove the delayed-ACK interaction.

For SYN queue limits: Increase the SYN backlog (`net.core.somaxconn`, `listen()` backlog, `net.ipv4.tcp_max_syn_backlog`) and enable SYN cookies; favour persistent, pooled connections for known institutional clients to reduce handshakes during the opening surge.

3. Recommended TCP Configuration Changes

- Set `TCP_NODELAY` on trading sockets to disable Nagle's algorithm — send small order messages immediately.
- Increase TCP receive/send buffer sizes and window scaling (`net.core.rmem_max` / `wmem_max`, `tcp_window_scaling`) to absorb order bursts.
- Increase the SYN backlog (`net.core.somaxconn`, `tcp_max_syn_backlog`) and enable SYN cookies to handle connection surges at market open.
- Enable TCP Fast Open (TFO) to cut handshake round-trips for frequently reconnecting clients.
- Use persistent, pre-warmed connection pooling instead of opening a new TCP connection per order.

- Tune or disable the delayed-ACK timer alongside TCP_NODELAY to prevent the two mechanisms compounding delay.

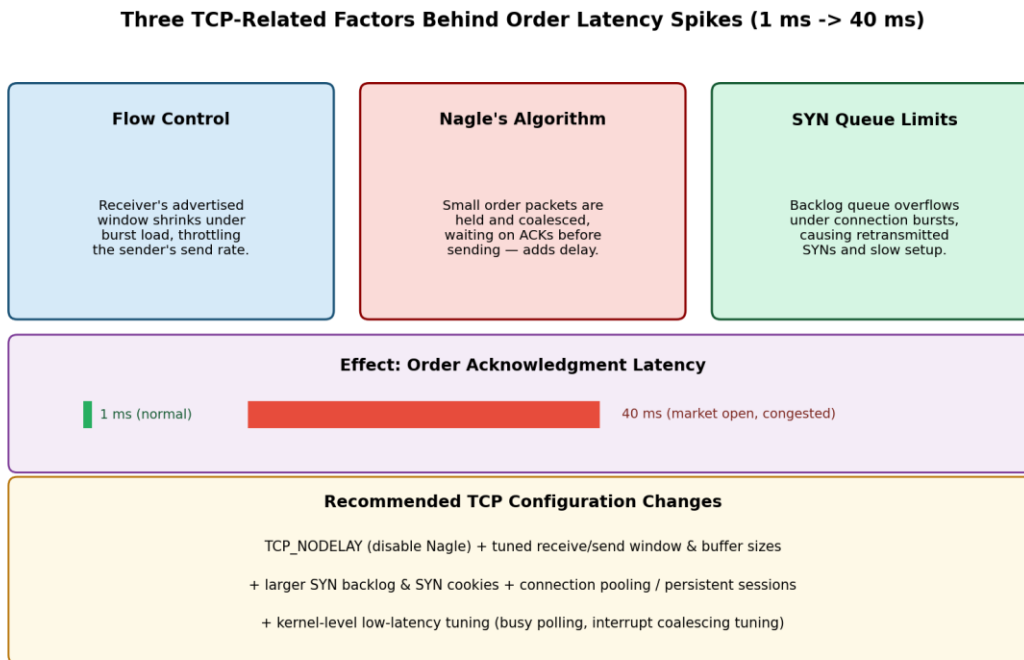


Figure 4.1:

Three TCP-layer factors compounding to raise order acknowledgment latency during market open.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand